The Relay

How signals travel along axons, cross synapses, and act on receptors

Rakesh Sengupta

School of Interwoven Arts and Sciences, Krea University

Introduction to Neuroscience · Sessions 10–12 (combined) · Reading: Bear et al., Ch. 4 (end), 5 and 6

Plan for today

Part	What we do	Time
Opening	From one spike to the next neuron; open Relay Lab	7 min
1 Along the axon	How a spike travels; myelin; MS (Lab Part 1)	25 min
2 Across the gap	Sparks or soups? The chemical synapse; adding up (Lab Part 2)	28 min
3 Keys and locks	Transmitters, receptors, drugs (Lab Part 3)	22 min
Wrap-up	Today on one page, again; exit ticket	8 min

You need: a laptop or phone (or a partner with one), the booklet, a pen. **Lost?** Booklet page 2 has today on one page.

Next class: Quiz 2 (Sessions 7–12).

How these slides work

The bold line at the top of each slide is the one thing to remember.

Labels at the top right:

IDEA one idea **STUDY** a real finding

THINK a question for you

RELAY LAB do this part of the lab

ANSWERS answers

Colours stay the same as before:

Na^+ sodium K^+ potassium

Cl^- chloride Ca^{2+} calcium

You can write instead of speaking, or pass.

Opening

From one spike to the next neuron

Opening | 1 Along the axon | 2 Across the gap | 3 Keys and locks | Wrap-up

you are here

Warm-up

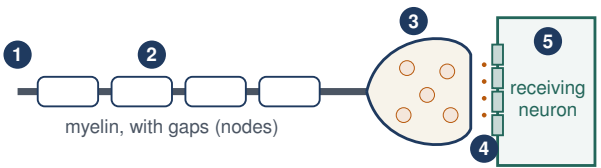
Think (writing or passing is fine)

You touch a hot pan. The message must travel about a metre up your arm, then pass from one neuron to the next, several times.

Where would you guess the time is lost: on the long trip along the wires, or at the junctions between them? Write a guess on booklet page 1.

Today on one page

One relay: along the axon, across the gap, into the next neuron.



1. **Along the axon:** each patch fires the next, like a fuse.
2. **Myelin** makes the spike jump from gap to gap: much faster.
3. **At the terminal:** Ca^{2+} rushes in; vesicles release transmitter.
4. **Across the gap:** transmitter binds to **receptors**.
5. **The receiving neuron** adds up its inputs and decides.

Open Relay Lab now

Keep the lab open beside the slides. Each lab slide tells you which part.

- Three short parts: **the race**, **adding up**, **keys and locks**. About 8 minutes each.
- Works on a laptop or a phone. Nobody sees your answers.
- **Nothing is saved or sent.**
- Press **Continue**. Each part is **locked**: its password appears on its lab slide. No rush.

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1 Along the axon

How a spike travels; myelin; multiple sclerosis

Opening | **1 Along the axon** | 2 Across the gap | 3 Keys and locks | Wrap-up

you are here

A spike travels like a burning fuse

Each patch of axon fires the next patch, so the spike arrives full size.

- **Na^+** rushing in at one patch spreads a little way inside the axon, pushing the next patch to threshold.
- The patch behind is refractory (Session 9), so the spike cannot turn back.
- A typical speed is about **10 m/s**, so one 2 ms spike is spread over about **2 cm** of axon (Bear et al., Ch. 4).

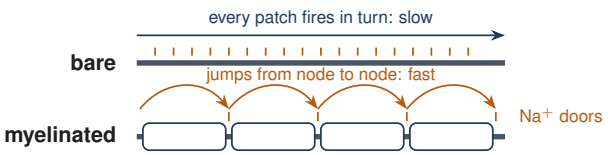
What sets the speed? Thickness

A wider axon lets the current spread farther ahead, so it is faster.

- Current ahead of the spike has two paths: **along** the inside of the axon, or **out** through the membrane. Bear compares this to water in a leaky garden hose.
- A wide axon carries more current along it, so the next patch reaches threshold sooner.
- The squid's escape axon is up to **1 mm** wide, so wide that it was first mistaken for a blood vessel (Bear et al., Ch. 4).

Myelin: a faster trick than thickness

Myelin insulates the axon, so the spike jumps from gap to gap.



(Bear et al., Ch. 4)

- Glia wrap the axon in layers of membrane: **myelin** (Session 6).
- Gaps called **nodes of Ranvier**, 0.2–2 mm apart, hold the **Na^+** doors.
- The spike leaps node to node: **saltatory** conduction (Latin *saltare*, to leap).

How fast is fast?

From walking pace to a racing car: about 1 to 120 m/s.

Axon	Example	Speed
thin, bare (about 1 µm)	slow, aching pain	about **1 m/s**
squid giant axon (0.5–1 mm, bare)	squid escape jet	about **20 m/s**
thick, myelinated (13–20 µm)	muscle sense, movement	**80–120 m/s**

A myelinated axon a fiftieth of the squid's width is several times faster. That is how a brain fits billions of fast axons into a skull. (Erlanger–Gasser fibre classes; Purves et al., *Neuroscience*)

When myelin is lost: multiple sclerosis

In MS, damaged myelin slows or blocks the spike.

- MS attacks myelin in the brain, spinal cord and optic nerves, at many sites; it often comes and goes over years.
- Symptoms: weakness, poor coordination, blurred vision, slurred speech.
- A test: show a checkerboard, and time the response over the visual cortex. In MS it arrives late (Bear et al., Box 4.5).

In Guillain–Barré syndrome, the immune system attacks myelin in the nerves of the body instead, slowing signals to the muscles.

Relay Lab, Part 1: the race

Time: 8 min | **Who:** alone or in pairs | **Where:** laptop or phone, booklet page 4

1. Type the **password** (right) and press **Unlock**.
2. Choose an axon and press **Send a spike**. It races 10 cm between two electrodes.
3. Race four: **1 µm bare**; **12 µm myelinated**; the **squid giant axon**; **12 µm myelinated with damaged myelin**.
4. Read each speed from the table; type it in and press **Check**.

PASSWORD
cheetah

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labs/relay-lab

You are done when

Four speeds are in the booklet and you answered the two questions.

Relay Lab, Part 1: answers

Myelin beats size. Damaged myelin slows the spike about sevenfold.

Axon	Time for 10 cm	Speed
1 µm, bare	100 ms	**1 m/s**
12 µm, myelinated	1.4 ms	**72 m/s**
squid giant axon (500 µm, bare)	4.5 ms	**22 m/s**
12 µm, damaged myelin patch	9.6 ms	**10 m/s**

At 1 m/s, a pain message takes about a second to travel a metre; at 72 m/s, about 14 ms.

So far

Spikes travel like a fuse; thickness and myelin make them fast.

A fuse

Each patch fires the next; never backwards.

Thicker, faster

Wide axons carry current farther ahead.

Myelin

The spike jumps node to node. Lost in MS.

Next: the spike reaches the end of the axon. How does it get to the next neuron?

2 Across the gap

Sparks or soups? The chemical synapse; adding up

The war of the soups and the sparks

For decades, scientists argued: is the message across a synapse electrical or chemical?

- **Sparks:** physiologists led by John Eccles: only electricity is fast enough.
- **Soups:** Henry Dale and Otto Loewi (the dream, from the interlude): chemicals.
- Karl Popper urged Eccles to make his theory testable enough to be **proved wrong**. In 1951, Eccles's own experiments did (Valenstein, 2005).

Think (writing or passing is fine)

Eccles switched sides, and won a Nobel Prize in 1963. Is changing your mind a weakness or a strength in science?

Two kinds of synapse

Electrical synapses pass current directly. Chemical synapses use a messenger.

Electrical (gap junction)

Cells only about **3 nm** apart, joined by protein pores. Current flows straight through: very fast, usually both ways. Common where neurons must fire together.

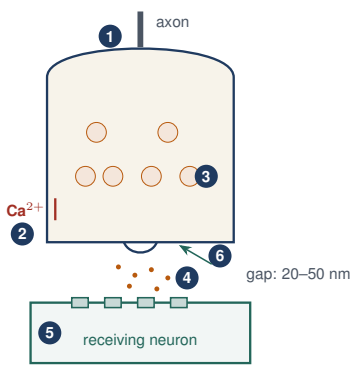
Chemical

A gap of **20–50 nm**. The sender releases a **neurotransmitter**; the receiver has **receptors**. One-way, slightly slower, but adjustable. Most brain synapses.

(Bear et al., Ch. 5)

A chemical synapse in six steps

Electrical signal → chemical → electrical again, in about a millisecond.



1. A spike arrives at the terminal.
2. **Ca^{2+}** doors open; **Ca^{2+}** rushes in.
3. Vesicles fuse and release transmitter.
4. Transmitter crosses the gap and binds to receptors.
5. Receptor channels open: an **EPSP** or **IPSP**.
6. Transmitter is cleared: reuptake or an enzyme.

(Bear et al., Ch. 5)

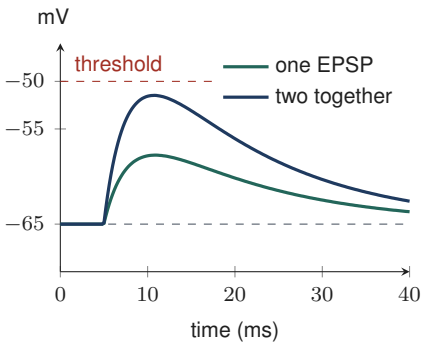
Messages come in packets

Transmitter is released in whole vesicles, called quanta.

- Bernard Katz found that responses come in steps, each the size of one vesicle's contents: a **quantum** of several thousand molecules.
- At a muscle, one spike releases about **200 vesicles**: a response of 40 mV or more, so the muscle always fires.
- At a typical brain synapse, often **one vesicle**: a response of a few tenths of a millivolt (Bear et al., Ch. 5).

Pushes up and pushes down

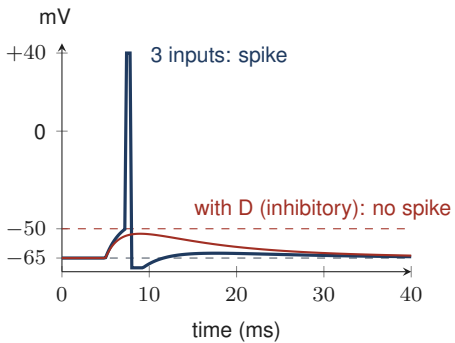
Excitatory inputs push towards threshold (EPSP). Inhibitory inputs pull away (IPSP).



- **Glutamate** usually excites: channels let positive ions in.
- **GABA** and **glycine** usually inhibit: Cl^- channels hold the voltage near rest.
- One input is rarely enough. The neuron must **add up** its inputs.

Adding up: space and time

Inputs add up if they arrive together (spatial) or close in time (temporal).



- **Spatial** summation: many synapses at once.
- **Temporal** summation: one synapse, firing again within about 1–15 ms (Bear et al., Ch. 5).
- An inhibitory input can veto the lot.

Relay Lab, Part 2: adding up

Time: 8 min | **Who:** alone or in pairs | **Where:** laptop or phone, booklet page 5

1. Type the **password** (right) and press **Unlock**.
2. Put the six steps of a chemical synapse in order.
3. Fire A alone, then A + B, then A + B + C. Then add D.
4. Fire A three times; find the longest gap that still makes a spike.

PASSWORD
chorus

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You are done when

The steps are in order, three numbers are checked, and two questions are answered.

Relay Lab, Part 2: answers

Three inputs together, or one input three times within 7 ms. Inhibition vetoes.

Rise from A alone	about **7 mV** (threshold is 15 mV above rest)
A + B together	about 13.5 mV: **still no spike**
Inputs needed together	**3**
A three times	spike up to a **7 ms** gap; none at 8 ms
A + B + C + D	**no spike**: inhibition

A brain neuron is a decision-maker, not a relay: it weighs thousands of inputs. A muscle synapse, by contrast, must work every time.

So far

At most synapses, a chemical carries the message, and the receiving neuron adds it up.

Soups won

Most synapses are chemical.

Six steps

Spike, **Ca**$^{2+}$, release, bind, open, clear.

Adding up

In space and time; inhibition can veto.

Next: which chemicals, which receptors, and how drugs get in on the act.

3 Keys and locks

Transmitters, receptors, drugs

Opening | 1 Along the axon | 2 Across the gap | **3 Keys and locks** | Wrap-up

you are here

Three families of transmitter

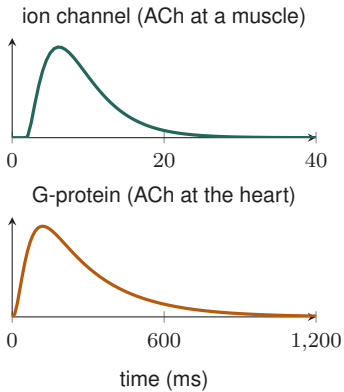
Neurotransmitters are amino acids, amines, or peptides.

Family	Examples	Typical job
Amino acids	**glutamate**, **GABA**, glycine	fast excitation or inhibition
Amines	**ACh**, dopamine, noradrenaline, serotonin	muscles; mood, attention, movement
Peptides	enkephalins (the brain's own opioids)	slow, long-lasting, e.g. pain

(Bear et al., Ch. 6)

Two kinds of lock

Ion-channel receptors are fast. G-protein receptors are slow, but amplify.



- **Ion channel:** the receptor *is* the door. Opens within a millisecond.
- **G-protein-coupled:** the receptor switches on a chain of molecules inside the cell. Slower, longer.
- One receptor can switch on 10–20 G-proteins: **amplification** (Bear et al., Ch. 6).

One key, many locks

The same transmitter can do opposite things, depending on the receptor.

- ACh at a skeletal muscle: **nicotinic** receptors (ion channels). The muscle contracts.
- ACh at the heart: **muscarinic** receptors (G-protein). The heart slows.
- This is **divergence**: every known transmitter acts on several receptor types (Bear et al., Ch. 6).

So “dopamine is the pleasure chemical” is too simple: what a transmitter does depends on where it lands.

How drugs act at a synapse

Drugs copy the key, block the lock, or keep the key in the lock longer.

Kind of drug	What it does	Examples
Agonist	copies the transmitter	nicotine (ACh receptors)
Antagonist	blocks the receptor	curare (ACh); caffeine (adenosine)
Reuptake blocker	keeps transmitter in the gap	fluoxetine/Prozac (serotonin)
Enzyme blocker	stops its breakdown	AChE inhibitors

(Bear et al., Ch. 5 and Table 6.1)

Relay Lab, Part 3: keys and locks

Time: 8 min | **Who:** alone or in pairs | **Where:** laptop or phone, booklet page 6

1. Type the **password** (right) and press **Unlock**.
2. Pick a synapse; press **Release transmitter** with no drug.
3. Try the agonist, the antagonist and the clearance blocker. Watch both panels.
4. Compare the muscle (ion channel) with a G-protein synapse.

PASSWORD
keyring

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You are done when

You tried all three drugs on two synapses and filled in the booklet table.

Relay Lab, Part 3: answers

Agonist: response even without release. Antagonist: tiny. Blocker: bigger, longer.

Drug	Muscle (ion channel)	Serotonin (G-protein)
none	100, about 8 ms	100, about 320 ms
agonist (nicotine)	already on; peak 150	already on; peak 160
antagonist (curare, caffeine)	15	15
clearance blocker (Prozac)	210, about 22 ms	290, about 590 ms

The ion-channel response is over in milliseconds; the G-protein response lasts hundreds. Same rules, different timescales.

Finding the brain's own morphine

Drugs revealed receptors, and receptors revealed the brain's own chemicals.

1. In the 1970s, Solomon Snyder and his student Candace Pert labelled opiate drugs with radioactivity.
2. The drugs stuck to specific sites on some neurons: **opiate receptors**.
3. Why would the brain have receptors for a poppy drug? The search found the brain's own opioids, the **enkephalins** and endorphins (Bear et al., Ch. 6 and Box 6.1).

Think (writing or passing is fine)

If the brain makes its own "morphine", what might it be for? (Hint: think of injuries during escape.)

So far

Which receptor a transmitter meets decides what it does; drugs hijack this.

Three families

Amino acids, amines, peptides.

Two locks

Ion channels fast; G-proteins slow, amplifying.

Drugs

Copy, block, or prolong the transmitter.

Wrap-up

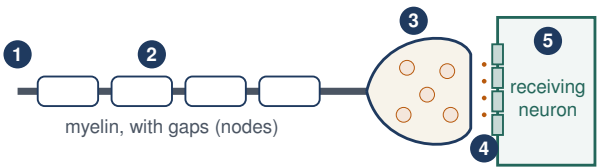
Today on one page, exit ticket and next time

Opening | 1 Along the axon | 2 Across the gap | 3 Keys and locks | **Wrap-up**

you are here

Today on one page, again

Can you explain each number to the person next to you?



1. **Along the axon:** each patch fires the next, like a fuse.
2. **Myelin** makes the spike jump from gap to gap: much faster.
3. **At the terminal:** Ca^{2+} rushes in; vesicles release transmitter.
4. **Across the gap:** transmitter binds to **receptors**.
5. **The receiving neuron** adds up its inputs and decides.

Three things to take away

1. A spike travels like a fuse. **Thick** and especially **myelinated** axons are fastest; MS damages myelin.
2. At most synapses, **Ca**$^{2+}$ triggers the release of a **chemical**; the receiving neuron **adds up** excitatory and inhibitory inputs.
3. The **receptor** decides the response: fast ion channels or slow G-proteins. Drugs copy, block or prolong transmitters.

Exit ticket and next time

Exit ticket: 3 minutes, booklet page 1

1. Why does myelin make a spike faster?
2. What does Ca^{2+} do at the axon terminal?
3. Name one agonist and one antagonist.

Next class

Quiz 2, on Sessions 7–12: membranes, the resting and action potentials, conduction, synapses, transmitters.

Revise with: the “today on one page” sheets, and the three labs.

Questions? rakesh.sengupta@krea.edu.in

References

Bear, M. F., Connors, B. W., & Paradiso, M. A. (2016). *Neuroscience: Exploring the Brain* (4th ed.), Chapters 4 (conduction, Box 4.5), 5 and 6.

Borck, C. (2017). Soups and sparks revisited: John Eccles' path from the war on electrical transmission to mental sparks. *Nuncius* 32, 286–329.

Brock, L. G., Coombs, J. S., & Eccles, J. C. (1952). The recording of potentials from motoneurones with an intracellular electrode. *J Physiol* 117, 431–460.

Hursh, J. B. (1939). Conduction velocity and diameter of nerve fibers. *Am J Physiol* 127, 131–139.

Pert, C. B., & Snyder, S. H. (1973). Opiate receptor: demonstration in nervous tissue. *Science* 179, 1011–1014.

Purves, D., et al. *Neuroscience* (Sinauer): fibre classes and conduction velocities.

Valenstein, E. S. (2005). *The War of the Soups and the Sparks*. Columbia University Press.

Relay Lab: Part 1 uses published rules of thumb for conduction speed; Part 2 a leaky integrate-and-fire neuron; Part 3 a simplified receptor model. Built with jsPsych 7 (de Leeuw, 2015, *Behav Res Methods* 47, 1–12). Plots on these slides come from the same models.

Check volume and page details before circulating.